

Computer Systems Architecture

CS1CA

Week 1 lecture 1 of autumn semester 2025

Lectured by

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Arrangements of the module

- It is a 20-credit module delivered in Semester 1, 2025.
- 24 lectures: listed below are topics being delivered.
 - General introduction to computer systems (1 lecture)
 - Information representation and binary operations (5 lectures)
 - Data manipulation (3 lectures)
 - Computer performance measures (1 lecture)
 - Instruction set architecture (4 lectures)
 - Processor and pipelining (4 lectures)
 - Memory hierarchy (4 lectures)
 - Storage and I/O systems (2 lectures)
- 2-hour revision lectures in Week 12
- 9-hour tutorials to enhance the learning
- 10-hour lab practicals (see timetable for the details)
- Two on-line tests in weeks 5 and 12 (50 mins each)

Assessment of the module

- Formative assessment
 - Exercises at the end of lecture notes
 - Lab tasks
- Summative assessment takes 30% of the module assessment.
 - Week 5 on-line test (15%)
 - Week 12 on-line test (15%)
- The final examination counts 70% of the module assessment.
 - It is a 3-hour examination in January/February 2026.

Blackboard

- **The Blackboard (BB)** is used as the main teaching and learning platform.
- **Teaching materials** are placed in the BB in weekly basis.
- **Announcement** of the BB is used for communication with students.
- **Module information** in the BB shows the detailed module schedules.
- **On-line tests** will take place in the BB.
- **Discission board**: place your questions there. You may answer others' questions, too.

Reference books

1. Computer Science: An Overview

J. Glenn Brookshear and Dennis Brylow, Pearson Education Ltd, 2020 (13th edition)

2. Computer Architecture: A Quantitative Approach

J L Hennessy and D A Patterson, Morgan Kaufmann Publishers, 2019 (6th edition)

3. Computer Organization and Design – The Hardware Software Interface

ARM Edition 2017 or

RISC-V Edition 2021

D A Patterson and J L Hennessy, Morgan Kaufmann

In this lecture

- General introduction to computer science and computer systems.
- Learning outcomes
 - To be able to give the definition of Computer Science
 - To be aware of various applications of computer science in different areas
 - To be able to briefly describe a computer system and interaction of hardware and software in computer system operations in terms of *von Neumann architecture*.

A brief history of computing

- The zeroth generation (1642-1945): mechanical computers
- The first generation (1945-1955): vacuum tubes
- The second generation (1955-1965): transistors
- The third generation (1965-1980): integrated circuits
- The fourth generation (1980-present): very large-scale integration (VLSI) => cloud computing, mobile computing ...
- The fifth generation (?): quantum computing, optical computing,
- Please explore more to find out potentials of future computer system development in aspects of computing power (*e.g.* cloud computing), miniaturization (*e.g.* smartphones), and connectivity (*e.g.* networks).

Computer science is the discipline that seeks to build a scientific foundation for such topics as computer design, computer programming, information processing, algorithmic solutions of problems, and the algorithmic process itself.

What is Computer Science?

- The answer or answers:/...../.....
- Computer Science (CS) is a diverse subject.
- It underpins almost everything we do in the modern world, even if we cannot see it on the surface. Examples?
- Building computer systems with regards to all related components, including software, hardware, networking, computer applications, *etc.*, is something you are going to learn in the course.

Questions to be answered

- What is a computer system?
- How does a computer system work?
- What are functions of hardware and software in a computer system?

We shall briefly answer the above questions in this lecture.

What is a computer system? -1

- A computer system includes **hardware**, **software**, and **peripheral devices** that are necessary to make the computer work for us.
- The coverage of computer systems could be from a smart phone to warehouse-scale computers.



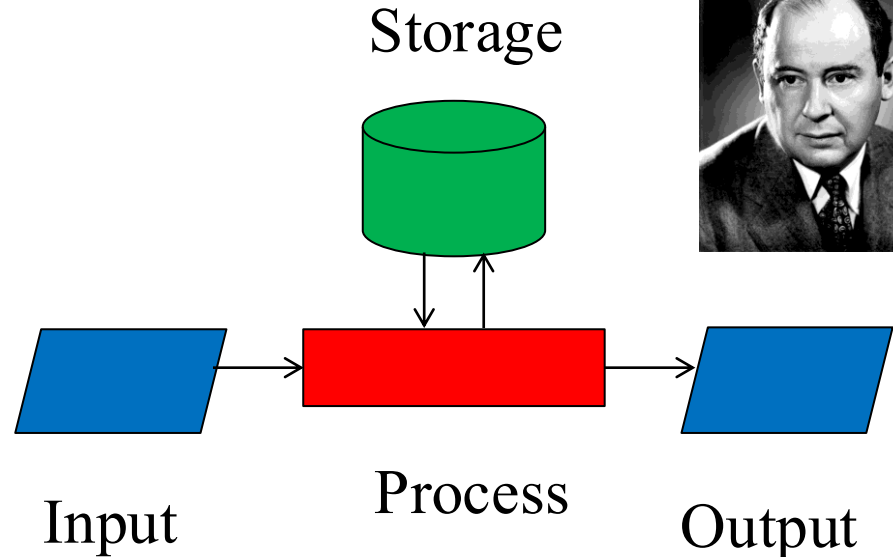
Smart phones



Google data-centre

What is a computer system? -2

- A stand-alone computer system consists of
 - Input devices
 - Processors
 - Output devices
 - Storage
- An *input-process-output* model is used for the work conducted by a computer.



von Neumann architecture
Stored program concept

Components of a computer system (1)

- **Computer hardware**: provide the physical mechanisms to input and output data, for manipulating data, and for electronically controlling various input, output, and storage.
- **Computer software** (including system software and application software): provide instructions that tell the hardware exactly what tasks are to be performed and in what order.

Components of a computer system (2)

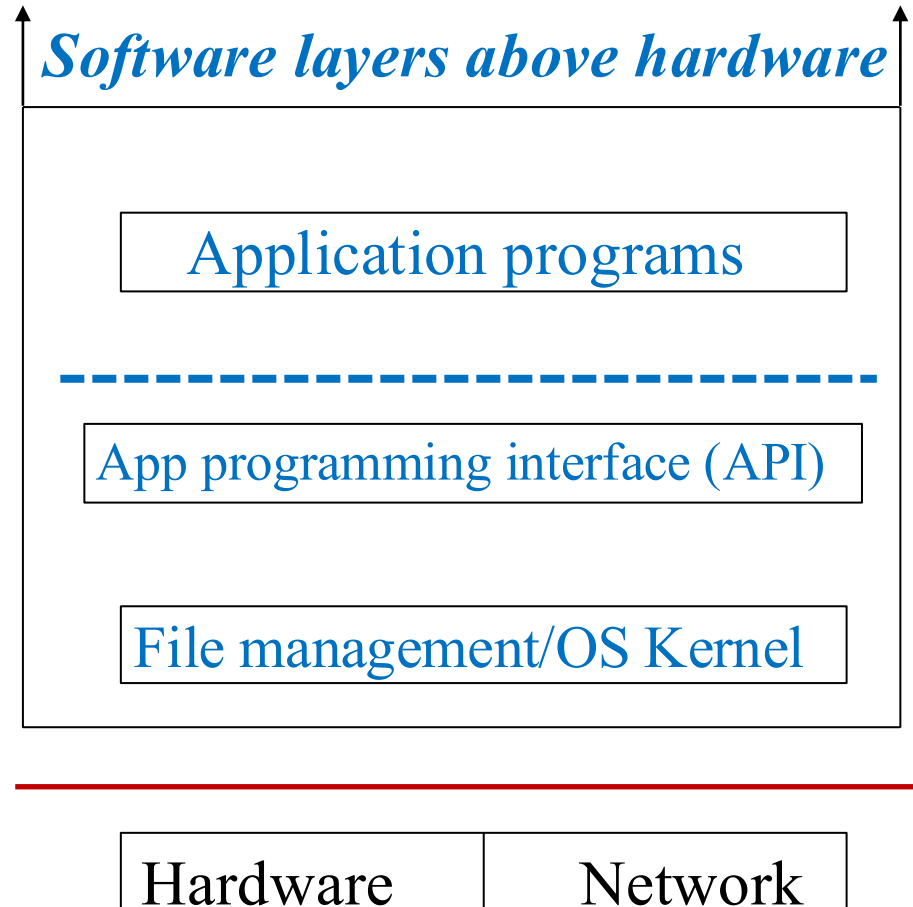
- **Data** are in different types, such as numeric, graphic, text, multimedia..., but in all cases it must be in a form that the computer can manipulate (binary representation).

For interconnected computer systems, there is

- **Communication component:** refers to hardware and software that transport programs and data between interconnected computer systems.

Hardware and software – joint efforts

- How does a computer system work?
- It needs joint efforts of hardware and software.
- Software provides instructions to tell hardware what to do.
- Hardware takes the instructions to conduct the tasks.

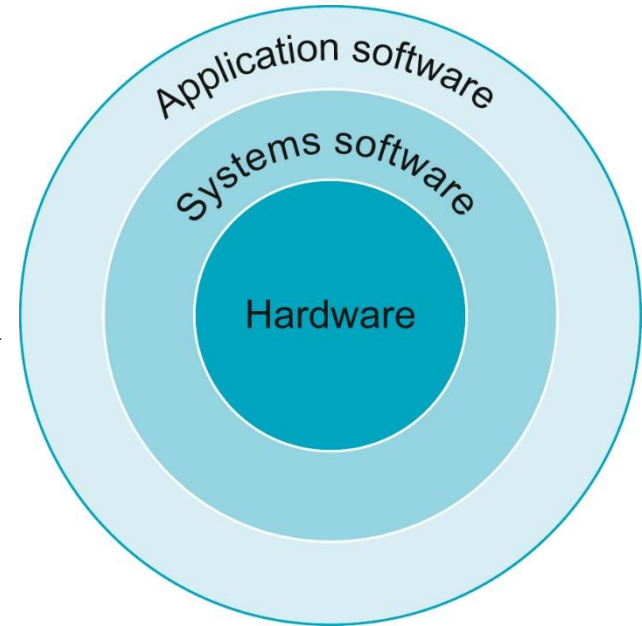


Computer hardware

- **Central Processing Unit (CPU)**: the CPU is viewed as a composition of three primary subunits, arithmetic/logic unit (ALU); control unit, and interface unit.
- **Internal memory**: **R**andom-**A**ccess **M**emory (RAM) and **R**ead-**O**nly **M**emory (ROM).
- **Storage**: hard drives (HDD and SSD), CD/DVD, USB flash stick, storage space in a server, ...
- **Input devices**: keyboard, mouse, storage units, ...
- **Output devices**: monitor, printer, speaker, storage units, ...
- **Buses**: physical connection that makes it possible to transfer data from one location to another in a computer system

Computer software

- Application software usually refers to a user's program which conducts a specific task.
- System software, *i.e.* **operating system**, interfaces between a user's program and hardware.
 - Handle basic input and output operations
 - Manage the resources of a computer
 - Provide protection schemes to the computer system
- **Compilers** are needed when programs are written in a high-level language, such as Java, Python, C/C++, etc.
 - Compiler translates a program written in a high-level programming language into **instructions** that the hardware can execute.



Summary of the introduction

- You will learn the afore-mentioned components involved in a computer system in this course.
- Computer science and technology are prevalent in the modern world.
- It is not always easy to predict the potential of applications of computer science, such as artificial intelligence (AI), which is closely related to computer science.
- Computer science and technology are still evolving. Hope you work hard and enjoy the study.

End of lecture 1: questions and exercises

- What is computer science?
- What is a computer system? Articulate the input-process-output model as *von Neumann architecture*.
- Briefly explain the four main components in a computer system.
- What is the difference between application software and system software?
- Why do we need an operating system in computer operations?
- What does a compiler do?



A website with useful resources, which may be interesting to you.

<http://computersciencestudent.com>

Questions?

